

MATHEMATICS

MATHEMATICS GRADE 12 SOURCE-EMBEDDED SMOKE TEST

SUBJECT	MATHEMATICS
GRADE	12
ASSESSMENT TASK	MATHEMATICS GRADE 12 SOURCE-EMBEDDED SMOKE TEST
MARKS	27
TIME	45 minutes

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.
2. Answer all questions or complete all activities as instructed.
3. Show all working where calculations, planning, diagrams or explanations are required.
4. Use the space provided by the educator, answer booklet or learner task book.
5. Write neatly and clearly.
6. Do not use a calculator unless the educator allows it.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Diagram Or Model

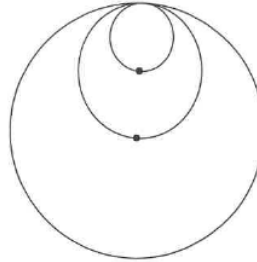
Mathematics/P1

4
NSC Confidential

DBE/November 2024

QUESTION 3

A circle with radius 6 cm is drawn.
A second, smaller circle is drawn through the centre of the first circle and also touches the first circle internally, as shown in the diagram.
A third, smaller circle is drawn through the centre of the second circle and touches the second circle internally. The process of drawing circles continues and forms a geometric pattern.



- 3.1 Write down the radius of the 3rd circle. (2)
- 3.2 Calculate the sum of the areas of the first 10 circles. (4)
- 3.3 Which circle has a diameter of $\frac{3}{128}$ cm? (4)
- [10]

QUESTION 4

Given: $f(x) = a^x - 1$ for $a > 0$. $B\left(2; -\frac{5}{9}\right)$ is a point on f .

- 4.1 Calculate the value of a . (2)
- 4.2 Write down the range of f . (1)
- 4.3 Sketch the graph of f . Clearly show the intercepts with the axes and asymptotes, if any. (3)
- 4.4 It is further given that C is a point on f at $y = \frac{19}{8}$.

Determine the coordinates of C' , the image of C, when C is reflected about the line $y = x$. (3)

[9]

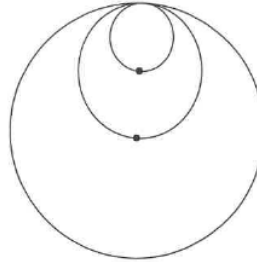
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SOURCE 2: Official-paper exemplar: Graph Or Chart

QUESTION 3

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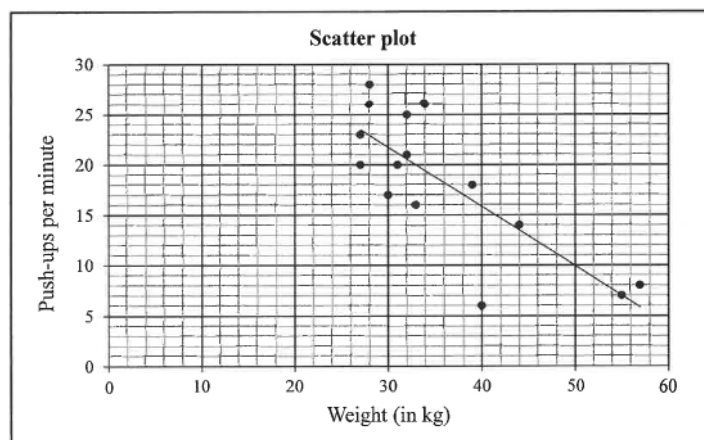
[9]

SOURCE 3: Official-paper exemplar: Data Table

QUESTION 1

At the beginning of a season, the coach of a junior boys' rugby team recorded the weight (in kg) of the 15 players in his team and the number of push-ups that each player was able to do in one minute. The data is represented in the table and scatter plot below. The least squares regression line for the data is drawn. ~

Weight (in kg) (x)	34	32	40	27	33	28	27	55	39	44	30	57	28	32	31
Number of push-ups per minute (y)	26	21	6	20	16	26	23	7	18	14	17	8	28	25	20



- 1.1 Determine the equation of the least squares regression line for the data. (3)
- 1.2 Write down the correlation coefficient. (1)
- 1.3 The coach uses the least squares regression line to set the target for the minimum number of push-ups by each team member according to their weight. Predict the number of push-ups that a member of the team, who weighs 29 kg, should do to meet the target. (2)
- 1.4 Write down the mean number of push-ups for the given data. (1)
- 1.5 The players trained hard during the season. At the end of the season, the coach reported that each player was able to do 5 more push-ups per minute than they did at the beginning of the season. How does the increase in the number of push-ups influence the standard deviation of the data? (1)
- 1.6 At the beginning of the season, the coach used the least squares regression line as the minimum target for a player to aim for. Determine the maximum possible increase in the number of push-ups that a team member must obtain to reach the minimum target. (2)

[10]

VISUAL MATERIAL

FIGURE 1: Working Grid And Method Marks

Mathematics		CALCULATION WORKSPACE
Mathematics Grade 12 Source-Embedded Smoke Test		
Working grid and method marks		
1.1 (4)	Refer to SOURCE 1. Identify one important feature of the diagram or model and explain how it could be used as evidence in Mathematics.	
1.2 (4)	Refer to SOURCE 2. Identify one important feature of the graph or chart and explain how it could be used as evidence in Mathematics.	
1.3 (4)	Refer to SOURCE 3. Identify one important feature of the data table and explain how it could be used as evidence in Mathematics.	
1.4 (5)	Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why.	
Award method marks from visible working before final-answer marks.		

QUESTIONS

QUESTION 1: SOURCE INTERPRETATION

Instructions: Use the supplied source material. Answer from visible evidence and relevant subject knowledge.

1.1. Refer to SOURCE 1. Identify one important feature of the diagram or model and explain how it could be used as evidence in Mathematics. **(4)**

1.2. Refer to SOURCE 2. Identify one important feature of the graph or chart and explain how it could be used as evidence in Mathematics. **(4)**

1.3. Refer to SOURCE 3. Identify one important feature of the data table and explain how it could be used as evidence in Mathematics. **(4)**

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

QUESTION 2: METHOD AND REVIEW

Instructions: Evaluate whether this generated assessment pack is ready for a real learner group.

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

MARKING GUIDELINE

TASK 1: SOURCE INTERPRETATION

1.1. Refer to SOURCE 1. Identify one important feature of the diagram or model and explain how it could be used as evidence in Mathematics. **(4)**

- Relevant feature from SOURCE 1 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.2. Refer to SOURCE 2. Identify one important feature of the graph or chart and explain how it could be used as evidence in Mathematics. **(4)**

- Relevant feature from SOURCE 2 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.3. Refer to SOURCE 3. Identify one important feature of the data table and explain how it could be used as evidence in Mathematics. **(4)**

- Relevant feature from SOURCE 3 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

- Two supplied sources are compared.
- A justified judgement is made.
- Reasoning mentions readability, data richness, visual clarity or subject fit.

TASK 2: METHOD AND REVIEW

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

- Check source readability and relevance.
- Check CAPS topic/grade alignment.
- Check mark allocation and memo correctness.
- Check copyright/reuse conditions for source material.

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

- They reveal actual source types used in exams.
- They reduce invented visual nonsense.
- They still require adaptation and educator approval.

RUBRIC

Criterion	Marks	Descriptor
Source evidence use	10	Answers use visible evidence from supplied sources.
Subject accuracy	10	Answers use correct subject concepts and terminology.
Review readiness	10	Risks and approval checks are identified clearly.

EDUCATOR REVIEW CHECKLIST

- Confirm each embedded source is readable after PDF export.
- Confirm source use/copyright status before client delivery.
- Confirm questions refer only to supplied or clearly stated evidence.
- Confirm memo and mark allocation match the subject route.